Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/21/2005
PI Name:	Kostelecky, V. Alan Ph.D.		
Project Title:	Theoretical Studies of Lorentz and CPT Symmetry		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kostelec@indiana.edu	**Fax:**	FY 812-855-5533
PI Organization Type:	UNIVERSITY	**Phone:**	812-855-1485
Organization Name:	Indiana University		
PI Address 1:	Physics Department, Indiana University		
PI Address 2:			
PI Web Page:	http://www.physics.indiana.edu/		
City:	Bloomington	**State:**	IN
Zip Code:	47405	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	01/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2914		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		
Task Description:	A general extension of the Standard Model of particle physics incorporating gravity, known as the Standard-Model Extension (SME), allows for the possibility of Lorentz and CPT violation while otherwise preserving conventional physics. Sensitive experiments searching for predicted effects include ones that may be advantageously performed in a microgravity environment. To develop and study theoretically possible violations of spacetime symmetry and to identify sensitive experimental tests for the associated effects. High-sensitivity tests of fundamental symmetries can shed light on physics at the Planck scale and could offer experimental support for an underlying theory such as strings.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	An analysis of data was performed from a clock-comparison experiment that yielded several stringent new limits on a number of coefficients for Lorentz and CPT violation in the SME. A comprehensive study of the neutrino sector of the SME was undertaken, suggesting possible signals in existing and future experiments with atmospheric neutrinos from cosmic rays, neutrinos from the Sun, and supernovae. The static limit of Lorentz-violating electrodynamics was studied and experiments to test the predicted effects were proposed. A comprehensive study of the gravitational sector of the SME was performed, including the Riemann spacetimes of Einstein's General Relativity and the generalized Riemann-Cartan spacetimes. This work has numerous potential implications for and applications to experiment.
Bibliography Type:	Description: (Last Updated: 03/05/2008)
Articles in Peer-reviewed Journals	F. Cane et al. "Bound on Lorentz and CPT violating boost effects for the neutron" Phys. Rev. Lett. 93, 230801 (2004) , Dec-2004
Articles in Peer-reviewed Journals	Q. Bailey and V.A. Kostelecky "Lorentz-violating electrostatics and magnetostatics" Phys. Rev. D 70, 076006 (2004) , Oct-2004
Articles in Peer-reviewed Journals	V.A. Kostelecky and M. Mewes "Lorentz violation and short-baseline experiments" Phys. Rev. D 70, 076002 (2004) , Oct-2004
Articles in Peer-reviewed Journals	V.A. Kostelecky and M. Mewes "Lorentz and CPT violation in the neutrino sector" Phys. Rev. D 70, 031902 (R) (2004) , Aug-2004
Articles in Peer-reviewed Journals	V.A. Kostelecky "Gravity, Lorentz violation, and the Standard Model" Phys. Rev. D 69, 105009 (2004) , May-2004
Articles in Peer-reviewed Journals	V.A. Kostelecky and M. Mewes "Lorentz and CPT violation in neutrinos" Phys. Rev. D 69, 016005 (2004) , Jan-2004